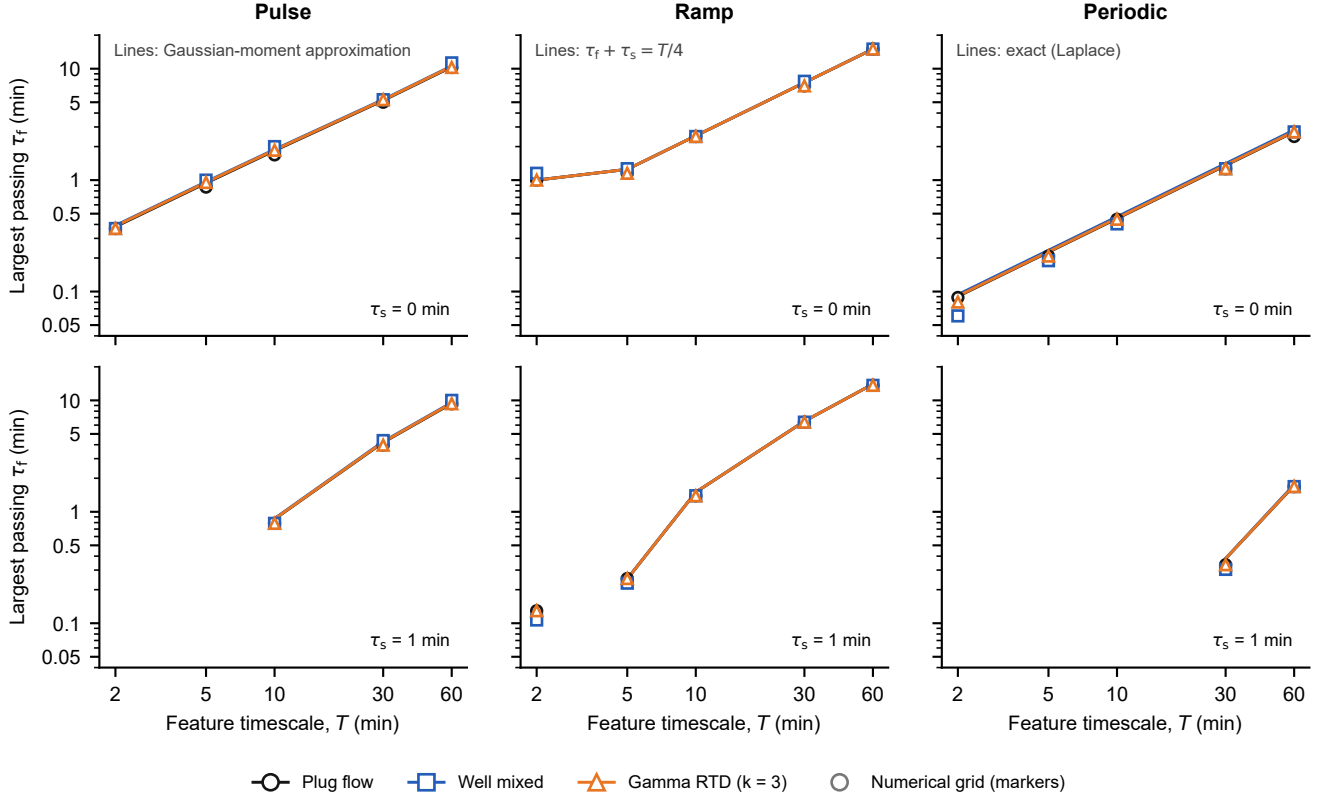

Pulse
Ramp
Periodic
Lines: Gaussian-moment approximation
Lines: $\tau_f + \tau_s = T/4$
Lines: exact (Laplace)
Largest passing τ_f (min)
τ_s = 0 min
τ_s = 1 min
10
5
1
0.5
0.1
0.05
2
5
10
30
60
Feature timescale, T (min)
Plug flow
Well mixed
Gamma RTD (k = 3)
Numerical grid (markers)